

Channel Sets Drift During Long Reasoning

Abstract

W4A16 long-reasoning quantization often protects a fixed set of high-risk activation channels. Channel-Set asks whether that static unit is stable enough to support a positive method. The current evidence supports a measurement/regime result rather than a confirmed adaptive method: top-channel sets drift substantially across reasoning horizons and model packets, with top-1% strict set-leaving of 0.634245 for Granite-Tiny, 0.566235 for Granite-Small, 0.533713 for Nemotron-3, and 0.670573 for the Phase 5 Transformer packet. Within-set shuffling is smaller but nontrivial. C-A1 remains parked because the cached gate path was contaminated and the fresh three-model same-row ParoQuant-vs-tight-clip matrix is missing. The paper contributes a scoped regime claim: static channel-set protection is a weak abstraction for these long-reasoning traces, and any future positive method must beat ParoQuant/static baselines under split-clean same-row native evidence.

Claim-Boundary Box

Supported	Unsupported	Future-only
Top-1% strict set-leaving is high across the four cached packets: 0.533713 to 0.670573.	A confirmed-positive claim is unsupported.	C-A1 can become a positive only after fresh split-clean native paired materialization.
Drift is not a single-threshold artifact across 0.5%, 1%, 2%, and 5% thresholds. C_U1, C_S1, and C_Y5 blockers are real schema/floor/native-pairing blockers.	C-A1 is not confirmed and does not beat ParoQuant in a claim-ready way. No held-out method confirmation is claimed.	A GPU/native systems card remains separate from this paper-finalization pass. OSC/DecDEC defenses can validate a future method after same-row pairing exists.

1. Introduction

Long reasoning changes which activation channels matter. A static W4A16 outlier-protection method can work only if the protected channel set remains stable enough across the reasoning trace. The Channel-Set campaign tested adaptive possibilities, but the confirmed result is simpler and sharper: the set

itself moves.

This matters for both method design and evaluation. If later high-risk channels leave the early protected set, then a static top-channel abstraction is incomplete. If an adaptive method appears to win without a paired ParoQuant/static baseline on the same rows, the win may be a cache artifact or a contaminated split artifact rather than a real method.

Contributions:

1. A long-reasoning channel-set drift measurement. At top 1%, strict set-leaving is above 0.53 for every reported packet (Figure 1).
2. A threshold sweep showing that the result is not a single chosen threshold (Figure 2).
3. A separation between strict set-leaving and within-set rank shuffling (Figure 3).
4. An audit-preserving regime map. C-A1 remains parked; C_U1, C_S1, and C_Y5 are blocked by missing schema, floor, or native-pairing requirements.

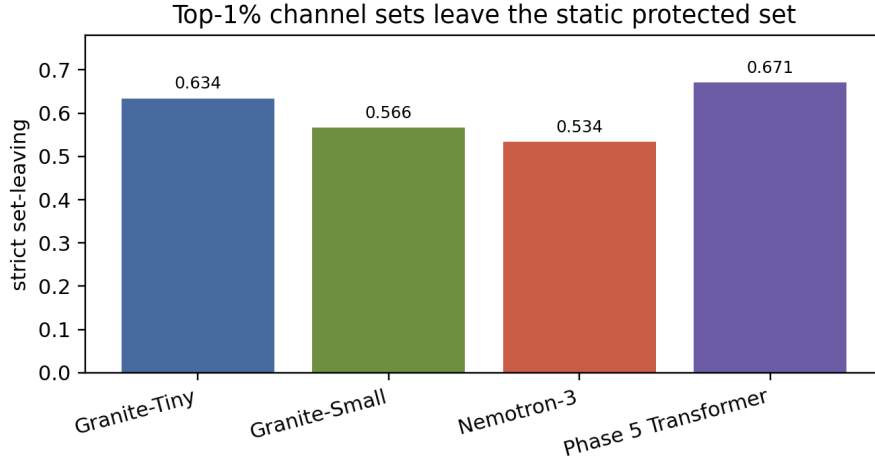


Figure 1: Top-1 set leaving

Figure 1: More than half of later high-risk top-1% channels leave the static protected set in every reported packet.

2. Measurement Setup

For a top-k channel threshold, define the protected set at one trace segment and ask how many later high-risk channels are outside that set. This is strict set-leaving. A value near zero would support static protection; a value above one half means the static set misses a large portion of later high-risk channels.

Within-set shuffling is measured separately: among channels that remain inside the protected set, their relative rank can still change. This affects allocation inside a fixed protected set but cannot recover channels that leave the set entirely.

The detailed provenance is in `tables/provenance.md`; the final regime checklist is in `tables/regime_checklist.md`.

3. Results

3.1 Top-1% Strict Set-Leaving Is Large

Packet	Strict set-leaving	Within-set shuffling
Granite-Tiny	0.634245	0.175260
Granite-Small	0.566235	0.270935
Nemotron-3	0.533713	0.269082
Phase 5 Transformer	0.670573	0.165737

The top-1% values all exceed 0.53, and the largest reaches 0.670573. Static protection is therefore not a sufficient abstraction for these traces.

3.2 The Result Is Stable Across Thresholds

Figure 2: Strict set-leaving remains high across the 0.5%, 1%, 2%, and 5% thresholds.

The threshold sweep reports all thresholds rather than selecting one post hoc. The exact values move, but the regime does not: strict set-leaving stays high across nearby thresholds for all four packets.

3.3 Strict Leaving Dominates Within-Set Shuffling

Figure 3: Within-set shuffling exists, but strict set-leaving is the larger effect.

Within-set shuffling at top 1% ranges from 0.165737 to 0.270935. That is nontrivial, but it is smaller than strict set-leaving. The main failure mode for

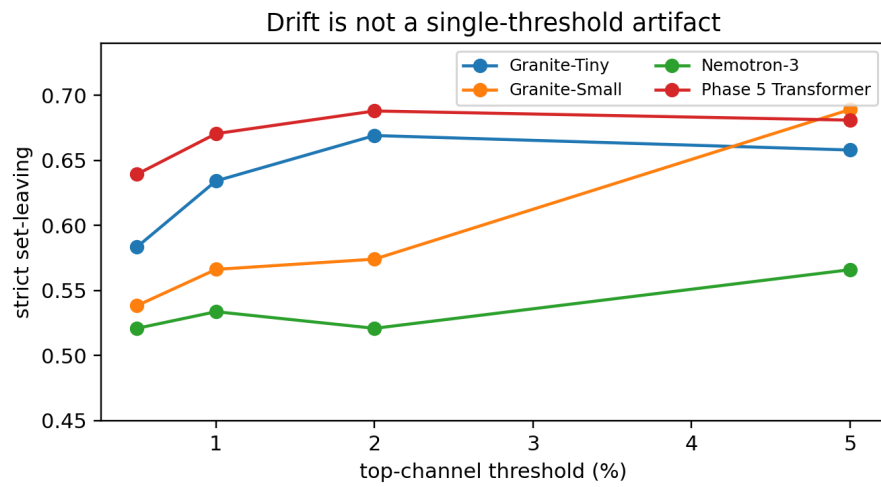


Figure 2: Threshold sensitivity

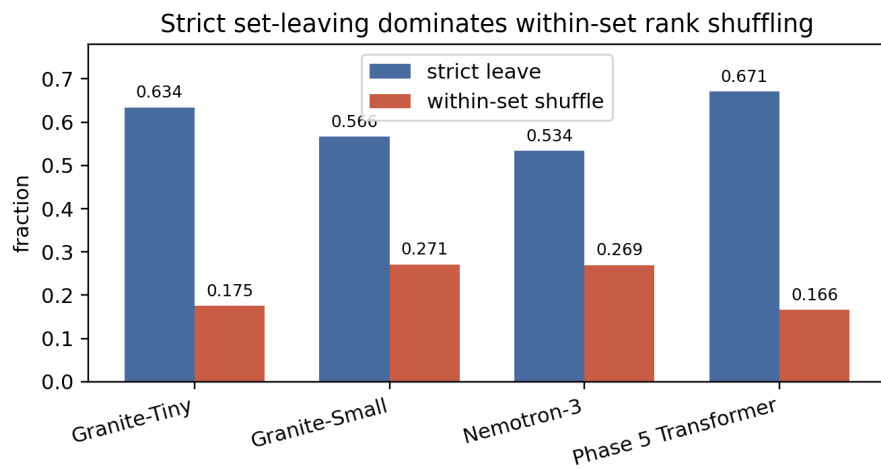


Figure 3: Within-set shuffling

a static set is not only that the protected channels change rank; many later high-risk channels are absent from the protected set at all.

4. Why C-A1 Is Parked, Not Confirmed

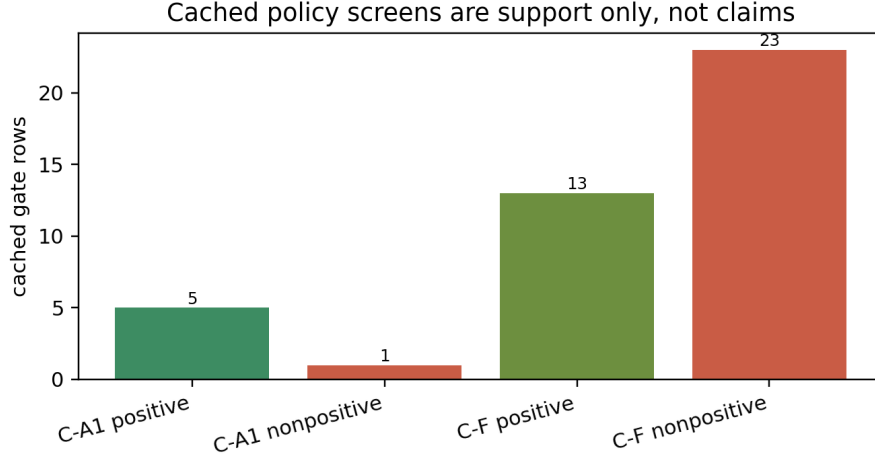


Figure 4: Cached policy screens

Figure 4: Cached C-A1 and C-F screens are support evidence only. They are not native same-row ParoQuant comparisons and cannot carry a positive claim.

C-A1 remains the nearest positive-method candidate, but it is not a paper claim. The cached screen has only 6 C-A1 gate rows, with 5 positive medians and 1 nonpositive median. That is enough to motivate a fresh materialization, not enough to claim a method.

Figure 5: The split-clean C-A1 sentinel matrix is incomplete. DeepSeek and Falcon have 12 same-row pairs, while Granite has 0 valid same-row pairs in the safe manifest.

The blocker is concrete. A future claim needs a Granite/DeepSeek/Falcon same-row matrix with ParoQuant baseline and tight-clip C-A1 outputs. The safe manifest has DeepSeek and Falcon same-row pairs, but Granite is missing/invalid. C-A1 remains parked until this is rebuilt.

5. Defenses And Schema Blockers

Figure 6: Defense and router ideas are blocked by missing labels, insufficient rows, or missing native pairing.

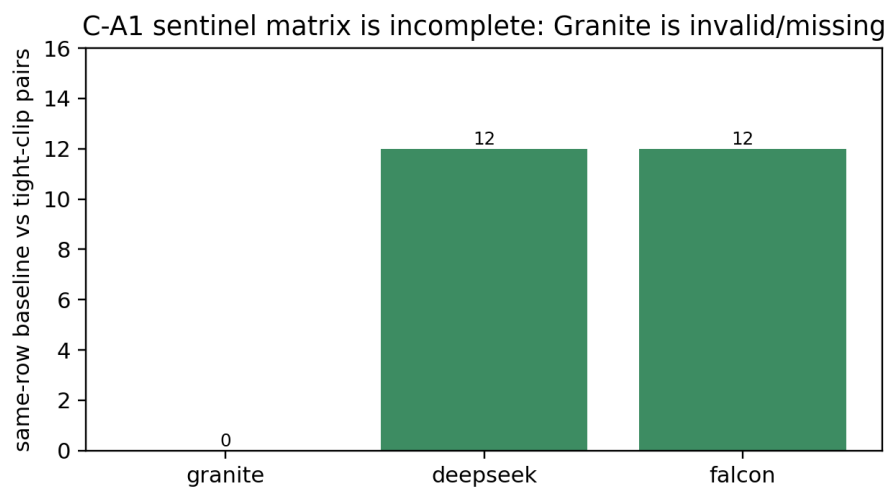


Figure 5: C-A1 sentinel status

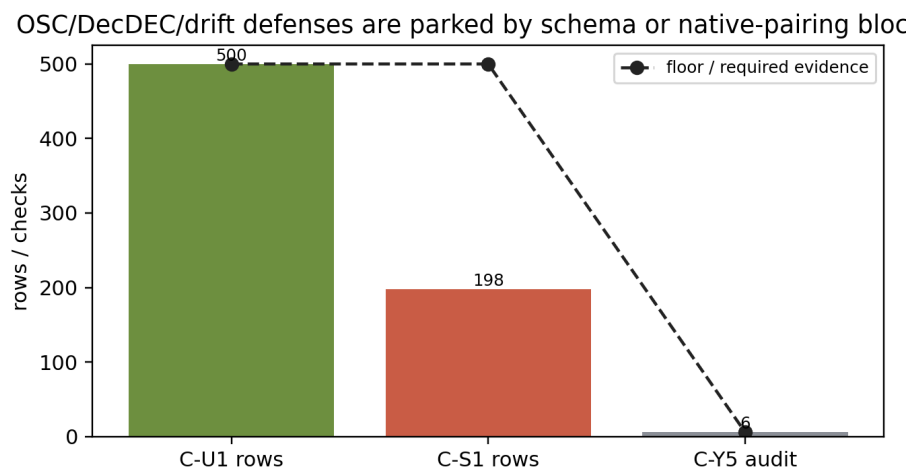


Figure 6: OSC DecDEC drift defense status

The defense stack is useful because it prevents overclaiming:

- C_U1 materializes 500 KL trajectory rows, but lacks paired difficulty/policy-uplift labels.
- C_S1 finds only 198 eligible rows against a 500 row floor. The floor is not lowered.
- C_Y5 has defense inputs and audit checks, but the three-model native C-A1-vs-ParoQuant packet is missing.

These are honest blockers, not negative proof against every future method. They define the evidence required before a positive Channel-Set method can be claimed.

6. Related Work

Channel-Set is closest to W4A16 quantization, outlier-channel protection, rotation-based quantization, and residual correction methods such as ParoQuant, OSC, DecDEC, ResQ, OSCAR, and OScaR. Those systems are the correct baseline family for any future adaptive method.

The distinguishing view here is the long-reasoning channel-set object itself. Rather than assuming a protected channel set and measuring final answer accuracy, this paper measures whether the protected set remains the same object over the reasoning trace.

7. Limitations

This is a measurement/regime paper, not a confirmed positive-method paper. C-A1 remains parked. The paper does not claim to beat ParoQuant, does not claim a native GPU result, and does not claim held-out method confirmation.

The measurements are cached decomposition evidence. They are strong enough to establish the static-set drift regime, but a method claim needs fresh split-clean native paired rows, model/policy hashes, no-gap denominators, and write-once per-trace evidence.

The claim is also not that static channel sets are always wrong. It is that, in these long-reasoning packets, strict set-leaving is large enough that static protection alone is a weak abstraction.

8. Conclusion

Channel-Set contributes a scoped but useful result: long-reasoning outlier channel sets drift. At top 1%, more than half of later high-risk channels leave the static protected set in every reported packet. That turns static W4A16 protection from a stable object into a fragile assumption. A positive method may still

emerge from C-A1 or related adaptive policies, but only after fresh split-clean native paired materialization and the full ParoQuant/audit gate.